

Day 02 Assignment - Understanding My Network

THEORY

1. What is a computer network?

- A computer network is an interconnection of two or more computers (or devices) for the sending and receiving of information.

2. What is the difference between a client and a server?

- A client requests for information or services from a server and a server provides the requested services to the client.

3. What is a LAN?

- LAN refers to Local Area Networks like a small office network or a home network where the devices are usually connected by using switches.

4. What is a WAN?

- WAN refers to Wide Area Networks which consists of a number of smaller networks (LAN), interconnected by using routers.

5. What is the purpose of a switch?

- Switches are used to connect the individual devices in a LAN so that they can send and receive information with each other.

6. What is the purpose of a router?

- Routers are used to connect the various networks within a WAN allowing the smaller networks (LAN) to communication and share information.

7. What is a firewall?

- A firewall is a security device that is installed before or after the router to filter the network traffic based on a list of configured rules.

8. What is an IP address?

- An IP address is a unique logical address given to devices to identify them on a network.

9. What is a MAC address?

- It is a physical address given to devices to enable hop-to-hop delivery on a network.

10. What is a default gateway?

- It is the IP address given to devices to use when they need to communicate with a device outside their network. Default gateway is usually the IP address of the router.

11. What is DNS?

- It is used to convert domain names (like www.google.com) into the IP address so that we don't have to memorise IP addresses of websites.

12. Why is networking knowledge important for cybersecurity?

- Most of the attacks and vulnerabilities are usually related to networks. To understand malicious behaviour, how data moves from point to point, etc, a security professional needs to constantly analyse IP addresses, ports, protocols, and so on. Hence, networking knowledge is important.

NETWORK INFORMATION

13. Write your computer's private IPv4 address

- 192.168.1.37

14. Write your default gateway.

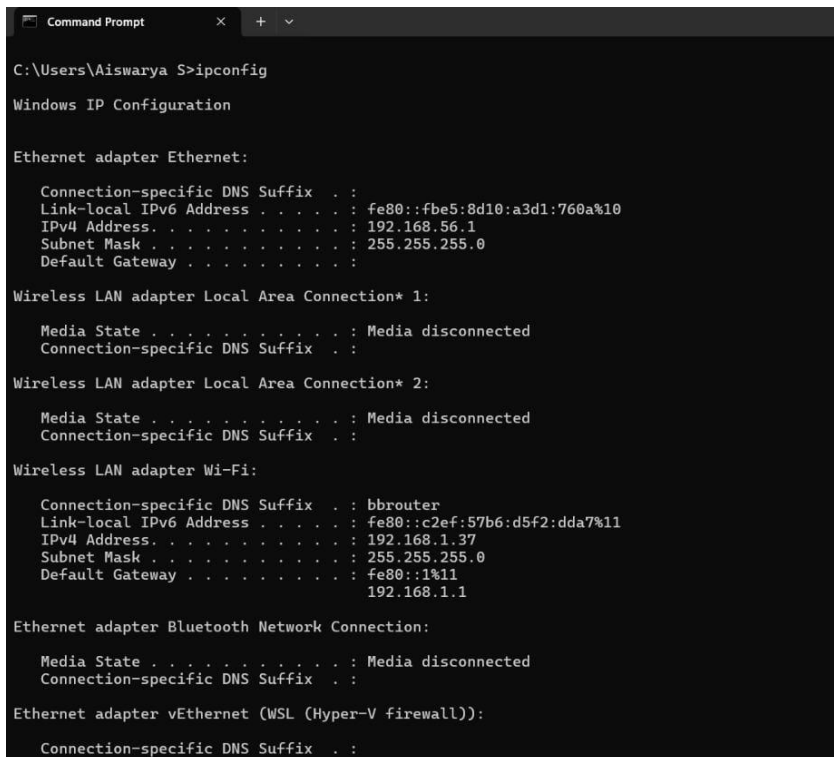
- 192.168.1.1

15. Write your DNS server.

- 192.168.1.1

SCREENSHOTS

- Ipconfig



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Command Prompt
C:\Users\Aiswarya S>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::f8e5:8d10:a3d1:760a%10
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : bbrouter
    Link-local IPv6 Address . . . . . : fe80::c2ef:57b6:d5f2:dda7%11
    IPv4 Address. . . . . : 192.168.1.37
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1%11
                                192.168.1.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

    Connection-specific DNS Suffix  . :
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Command Prompt
Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::f8e5:8d10:a3d1:760a%10
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : bbrouter
    Link-local IPv6 Address . . . . . : fe80::c2ef:57b6:d5f2:dda7%11
    IPv4 Address. . . . . : 192.168.1.37
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1%11
                               192.168.1.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::5c2d:ea81:ffe4:d02f%36
    IPv4 Address. . . . . : 172.25.176.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . : 

C:\Users\Aiswarya S>
```

- ping

```
C:\Users\Aiswarya S>ping google.com

Pinging google.com [142.250.77.142] with 32 bytes of data:
Reply from 142.250.77.142: bytes=32 time=23ms TTL=118
Reply from 142.250.77.142: bytes=32 time=24ms TTL=118
Reply from 142.250.77.142: bytes=32 time=23ms TTL=118
Reply from 142.250.77.142: bytes=32 time=23ms TTL=118

Ping statistics for 142.250.77.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 24ms, Average = 23ms

C:\Users\Aiswarya S>
```

- tracert

```
C:\Users\Aiswarya S>tracert google.com

Tracing route to google.com [142.250.77.142]
over a maximum of 30 hops:

  0  4 ms   2 ms   13 ms  192.168.1.1
  1  5 ms   4 ms   5 ms   100.73.0.1
  2  *      *      *      Request timed out.
  3  6 ms   5 ms   8 ms   41.151.164.202.asianet.co.in [202.164.151.41]
  4  29 ms  62 ms  *      194.148.164.202.asianet.co.in [202.164.148.194]
  5  77 ms  51 ms  23 ms  154.232.88.202.asianet.co.in [202.88.232.154]
  6  *      51 ms  29 ms  21.252.88.202.asianet.co.in [202.88.252.21]
  7  23 ms  26 ms  28 ms  192.178.121.39
  8  24 ms  23 ms  22 ms  142.251.55.63
  9  24 ms  25 ms  22 ms  lcmaaa-au-in-f14.1e100.net [142.250.77.142]

Trace complete.

C:\Users\Aiswarya S>
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